

Job Description for Atlas Research Engineer, SLAM & Spatial AI

Job Description

As a **SLAM & Spatial AI Research Engineer** on the **Atlas VLA Research team**, you will build the perception and geometric reasoning systems that give Atlas a grounded 3D understanding of the world. Your work spans the full spectrum from real-time SLAM and state estimation on humanoid hardware to offline reconstruction pipelines that produce the geometric scene structure used to train and condition large VLM/VLA models.

You will design real-time SLAM and perception-based state estimation that runs on Atlas, develop offline 3D reconstruction pipelines that turn teleop and robot logs into high-fidelity geometric data, and pursue research in spatial AI, grounding language and vision into 3D geometry so that learned policies can reason about space, not just pixels. You'll collaborate closely with perception, robotics, ML & system software specialists and rapidly test your work on state-of-the-art hardware.

How You Will Make an Impact:

- Design and implement real-time SLAM and perception-based state estimation for a mobile humanoid or specialized data collection devices operating in unstructured, dynamic environments
- Build offline 3D reconstruction pipelines (multi-view geometry, SfM/MVS, neural reconstruction, depth/pose fusion) that generate geometric scene structure to inform and supervise large VLM/VLA training
- Pioneer research integrating large VLA and VLM models with 3D spatial perception to enable semantic, language-grounded scene reasoning.
- Bridge classical geometric methods and learned approaches - knowing when to use optimization-based estimation versus learned representations, and how to combine them.
- Write high-quality, maintainable C++ and Python code that fits into a large production codebase.

We're Looking For:

- PhD in Robotics, Computer Vision, Machine Learning, Computer Science, or related fields (or equivalent experience).
- Prior experience building, and deploying SLAM, visual odometry, or 3D reconstruction systems for robots or autonomous vehicles.
- Strong background in one or more of the following:
 - Real-time SLAM, visual-inertial odometry, and state estimation
 - 3D reconstruction (SfM, MVS, multi-view geometry, neural/implicit reconstruction)
 - Probabilistic state estimation and sensor fusion (factor graphs, filtering, optimization on manifolds)
 - Spatial representations, grounding language/vision into 3D geometry, geometric foundation models
- Solid foundation in the math underlying geometric perception (Lie groups, nonlinear optimization, multi-view geometry).
- Strong analytical and debugging skills; ability to write reliable, well-structured research code in C++ and Python.

Nice to Have:

- Experience with modern ML frameworks (PyTorch, JAX) and an understanding of how perception outputs feed large-scale model training.
- Experience building reconstruction or data pipelines that produce training data for large vision or VLA models.
- Familiarity with VLA / large behavior models and how spatial grounding improves manipulation and long-horizon behavior.
- Publications in top-tier computer vision, ML, or robotics conferences (e.g., CVPR, ICCV, ECCV, RSS, ICRA, CoRL).

Why Join Us:

- Direct access to the world's most advanced humanoid robot: test your models on hardware quickly and often.
- A collaborative, inclusive team that values diverse perspectives and identities.
- The opportunity to do applied spatial perception and VLA research with real-world impact.
- A mission-focused environment where your work will define the future of general-purpose humanoids.

The base pay range for this position is between \$177,000 to \$225,000 annually. Base pay will depend on multiple individualized factors including, but not limited to internal equity, job related knowledge, skills and experience. This range represents a good faith estimate of compensation at the time of posting. Boston Dynamics offers a generous Benefits package including medical, dental vision, 401(k), paid time off and a annual bonus structure. Additional details regarding these benefit plans will be provided if an employee receives an offer for employment.

